

Trajectory Segmentation Using Oblique Envelopes

Sudheer Chelluboina
Computer Science and Engineering
Southern Methodist University
Dallas, TeX 75206, USA
schelluboi@smu.edu

Michael Hahsler
Engineering Management, Information, and
Systems
Southern Methodist University
Dallas, TX 75206, USA
mhahsler@smu.edu

ABSTRACT

Trajectory segmentation, i.e., breaking the trajectory into sub-trajectories, is a fundamental task needed for many applications dealing with moving objects. Several methods for trajectory segmentation, e.g., based on minimum description length (MDL), have been proposed. In this paper, we develop a novel technique for trajectory segmentation which created a series of oblique envelopes to partition the trajectory into sub-trajectories. Experiments with the new algorithm on hurricane trajectory data, taxi GPS data and simulated data of tracks show that oblique envelopes outperform MDL-based trajectory segmentation.

Categories and Subject Descriptors

H.2.8 [Database Management]: Database Applications - Data Mining

General Terms

Algorithms

Keywords

Trajectory Segmentation, Spatial-Temporal Data, GPS

1. INTRODUCTION

Usage of Location Based Services (LBS) is increasing significantly in day-to-day life [9]. LBS can answer the following questions: find a gas stations near a given location, or find the fastest route while avoiding road tolls. LBS systems mainly concentrate on positioning techniques and recording and tracking moving objects. LBS systems need to store, update and query locations/positions from large amounts of data generated by continuously moving objects, which leads to new challenges in the field of databases. With tracking methods like the Global-Positioning System (GPS), large amount of spatio-temporal data can now be easily obtained. This leads to increased need for data analysis in fleet management, traffic management, mobile communication, and

environment monitoring systems. Aims include to extract significant places for an individual or to predict the future movement of an individual.

Moving object analysis has become the most popular research interest in the area of GIS (Geographic Information Systems). An interesting point in moving object analysis is that the moving object is not considered as important as its location, so these moving objects are represented as the moving points whose trajectories are studied and analyzed to extract information. Analysis of trajectory data create opportunities for discovering behavioral patterns which can be exploited in various applications.

Segmenting trajectories is used as a preliminary step for various applications. Ecologists study the recurrent patterns and behavioral changes in animal tracking data. In this analysis the trajectory is split and annotated depending on animal movement. Usually this task is performed manually, but it can be automated using partitioning algorithm on trajectory segments by giving a formal definition for different animal movements, i.e., birds movement activity as soaring, circling [8] Meteorologists research is focused on improving forecast models for the location and intensity of hurricane landfall. Powel and Abernson [13] has shown how hurricane landfall forecasts improve by discovering similar sub-trajectories. Buchin et al. [3] define the trajectory segmentation problem as breaking a trajectory into the minimal number of segments (sub-trajectories) while still satisfying a given criteria (e.g., the deviation of a point from the segment's model). From the analysis of these sub-trajectories, we can obtain the information regarding the movement characteristics such as speed and group patterns; that is, we are trying to understand the behavior of moving objects along the trajectory.

In this paper we propose a new method for trajectory segmentation using the formation of sequential oblique envelopes (OE). Using hurricane data, taxi data and simulated data, we evaluate our trajectory segmentation method and compare the results with an existing algorithm based on MDL approach. Our results show that, our proposed algorithm yields a better approximation. The rest of the paper is organized as follows. Section 2 discusses some related work. In Section 3 we define our method and proposes the concept of sequential oblique envelopes. We present results of experimental evaluation in Section 4. Section 5 concludes the paper.

2. RELATED WORK

Trajectory data mining covers many topics which includes

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frequent pattern mining, trajectory clustering and trajectory outlier detection. Frequent pattern mining can be used to extract sets of moving objects which are moving parallel [7]. Flock patterns help to analyze the relative motion of a set of objects traveling within some specified range [10]. Lee et al. [11] proposed TRACCLUS, a partition-and-group framework for clustering trajectories which is the discovery of similar sub-trajectories, based on a partitioning algorithm that uses the minimum description length principle (MDL) [11]. The MDL principle has two components: $L(H)$ and $L(D|H)$ [9]. $L(H)$ is the length of the description of the hypothesis H (in bits); and $L(D|H)$ is the length of the description of the data when encoded with the help of the hypothesis (in bits). The set of trajectory segments correlates to H and the trajectory itself represents D . That is, $L(H)$ measures the conciseness and $L(D|H)$ measures the preciseness of the found trajectory segmentation. To minimize the sum of $L(H)$ and $L(D|H)$ represents a trade-off between the preciseness and conciseness. We re-implemented the approximate trajectory partitioning algorithm based on MDL approximation. It mainly has two computations, based on those computations the length of the candidate trajectory partition increases. That is MDL_{par} and MDL_{nopar} are computed for every point in the trajectory. MDL_{par} is the cost of a trajectory between points p_i and p_j ($i < j$) that is those points are the characteristic points. And MDL_{nopar} represents the MDL cost when assuming that there is no characteristic point between p_i and p_j that is when maintaining the original trajectory. If MDL_{par} value is greater when compared to MDL_{nopar} value than at that point the trajectory is partitioned otherwise the length of trajectory partition increases. It has a single parameter α to choose the roughness and smoothness of the track. That is the parameter α is multiplied with MDL_{par} such that it can accommodate more points between the characteristic points. We compare the algorithm to MDL because it is the best known and most widely used of the segmentation for outlier detection. Zhang et al. [18] first exploited the concept of micro clusters, that is the group of data points which are close to each other are likely to belong to one cluster. It was further investigated to not only find the group of objects that are close in that interval but also to find the objects that are moving together for some time interval [12]. These all proposed methodologies are compared with the MDL approximation method, so we also considered this method to compare with our new segmentation method. The CMON framework [4] proposes to deal with the problem of clustering moving objects. Chen et al. [5] proposed a new distance function for edit distance on real-valued sequence to measure the similarity between sub-trajectories.

Many trajectory data mining approaches use segmentation as a basic building block. Segmentation is defined as the partitioning of data into groups of individuals that are similar in a specific way. Segmentation is a common problem in time series analysis [17] and has been studied in trajectory mining for understanding the geographic trajectories of moving objects, such as vehicles and animals.

3. FORMATION OF SEQUENTIAL OBLIQUE ENVELOPES

Definition 1: (Discrete Trajectory) A discrete trajectory is defined as the sequence of points $\langle p_1, p_2, p_3, \dots, p_n \rangle$ where

$n \geq 1$ and each point p_i is defined by a triple (l_{lat}, l_{lon}, t) , where l_{lat} is the latitude, l_{lon} is the longitude and t is the time stamp at that location.

Definition 2: (Sub-Trajectory) A sub-trajectory is defined as the sequence of continuous points in a certain trajectory. That is in trajectory T and sub-trajectory T_{sub} is given by $\langle p_k, p_{(k+1)}, p_{(k+2)}, \dots, p_{(k+l-1)} \rangle$ where $1 \leq k \leq (n-1)$ and k is the starting point and n is the length.

Definition 3: (Trajectory Segmentation) A Trajectory Segmentation is the process of partitioning a trajectory into a set of sub-trajectories, where each sub-trajectory can be divided by a simple models (example: a straight line or a segment).

Definition 4: (Oblique Envelope (OE)) An oblique envelope (OE) is the smallest enclosing rectangle for all the data points of a sub-trajectory. It is given by $\langle start - point, width, end - point \rangle$.

For a sub-trajectory containing only a single point, the envelope would have side lengths of zero and once we increase the number of points in the sub-trajectory, then the lengths of the sides start to grow. We constrain the length of the shorter side of the rectangle to a user-defined threshold w and we try to find the minimal number of envelopes that can contain all data points in the trajectory. The envelopes then represent the trajectory segments.

The idea of the algorithm is to add data points to the sub-trajectory as long as the envelope can be enlarged to fit the point. Once a new point cannot be accommodated by the envelope, a new envelope is created. An example is given in Figure 1. The envelope is represented as a rectangle and the red color dot are the GPS data points. Here the first point will create the envelope around the data point with a user defined width w (shown in Figure 1(a)). For the next data point, the data point will be checked if the envelope can accommodate the new data point. In Figure 1(b) we see that the second data point can also be accommodated by enlarging the envelope in one direction. Figure 1(c) shows more data points that can be accommodated by the same envelope.

An oblique envelope can be defined by the two end points, say A and B and the user-defined width w . The angle of the envelope in the coordinate system is given by θ .

The algorithm to create sequential OEs is presented in Algorithm 1. With the first two data points, the first OE is created around the data points with the given width w . From the third data point, OE is created between first and third data point, then the data point between the first and third data points are checked whether it falls outside the OE. In computational geometry there are many strategies like Ray Casting, winding number to find the point inside the polygon or not. In most of the databases like Oracle-Spatial, MySQL and PostgreSQL, there is a inbuilt function called point-in-polygon to find the point lies inside the polygon. Here OE is considered as polygon, so we can use any of these functions to check the point falls inside the OE. If all points are in the OE then the new OE is kept as the current OE, otherwise the current OE is added to the list of OEs S and a new OE is started with the last two points.

Our algorithm has the nice property that it requires only one parameter w . A large w value will accommodate many data points and generate a smaller number of segments, whereas a small w value has the opposite effect.

4. EXPERIMENTS

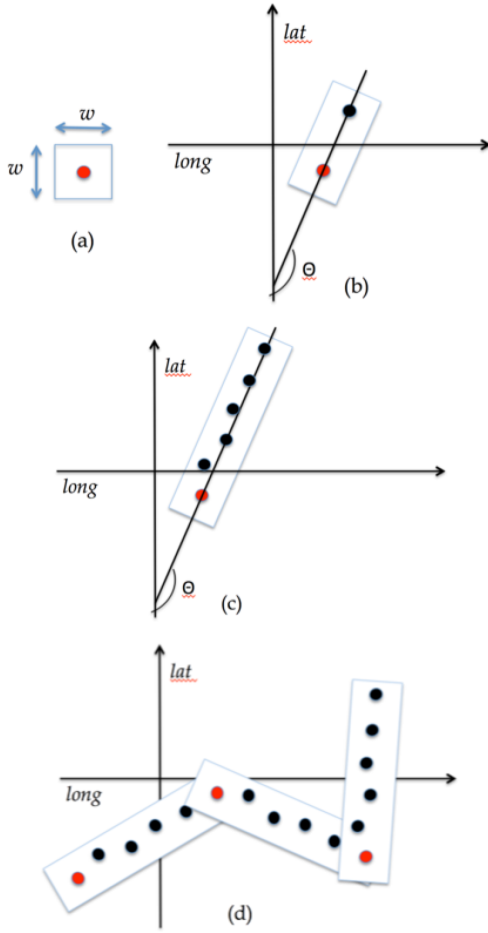


Figure 1: Evolution of Oblique Envelope

We use three datasets for experiments, two real-world trajectory dataset called hurricane dataset [14] and the taxi dataset [16, 15], and a simulated dataset. The hurricane dataset contains track information for all named hurricanes in the atlantic basin from 1982 to 2012. It contains 420 hurricanes with 9926 data points. A data point is recorded for every six hours. Figure 2 shows the segmentation results for a three storms. The blue dots represent the actual track points, and the red rectangles represent the computed OE.

Now we compare the parameter sensitivity with the existing algorithm called the partition-and-detect framework [11]. This algorithm uses minimal description length (MDL) principle for segmenting the trajectory. As mentioned in related work length of the trajectory partition is determined by the alpha value. Here we compare the parameter sensitivity between our algorithm parameter w and MDL algorithm parameter α .

To compare we use 10 sample trajectories from the taxi dataset. Taxi dataset is shown in Figure 4. Figure 3 shows that the number of segments depends on the algorithm's parameter. Both have a similar behavior and reasonable values for α are between 0.35 and 0.6, and for the distance parameter w between 2 and 200 meters. However, the advantage of w over α is that it directly corresponds to the maximal deviation a point from the general direction of the segment

Algorithm 1 Formation of Sequential OEs

Require: trajectory $T = \langle p_1, p_2, \dots, p_n \rangle$ and width w
Ensure: list of sequential OEs S

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1: current  $\leftarrow$  createOE( $p_1, p_2, w$ )
2: for each new data point  $p_i$  ( $i > 2$ ) do
3:   temp  $\leftarrow$  createOE(current.FirstDataPoint,  $p_i, w$ )
4:   for each data point  $p$  in current do
5:     if linside(temp,  $p$ ) then
6:       createNewOE  $\leftarrow$  true
7:       break
8:     end if
9:   end for
10:  if createNewOE = true then
11:    add current to  $S$ 
12:    current  $\leftarrow$  createOE(current.LastDataPoint,  $p_i, w$ )
13:  else
14:    current  $\leftarrow$  temp
15:  end if
16: end for
17: add current to  $S$ 

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can have.

Next we compare segmentation quality with taxi data and simulated data using randomly chosen way points and linear interpolation between the way points using GPSBabel [1]. GPSBabel is a package which translates between the GPS receivers and mapping programs like Google Earth. It is useful for manipulation, filtering duplicate points and also provides outputs in different formats as required by the user. Here hurricane data is not considered as each hurricane has only 15 data points, so for evaluation of quality it will provide insufficient data points for building and testing the model. GPS determines the location based on the following information. The precise locations of satellites are downloaded and the information is stored in the memory for future use. GPS receiver, calculates the distance between the satellite and the receiver using the signal time to reach the receiver from the satellite. Triangulation methods are used to calculate the approximate position when it receives signals from at least three satellites. The position determined by the GPS receiver is subject to several sources of error: multipath interference, satellite and receiver clock errors, orbit errors, satellite geometry, atmospheric interference, and selective availability. Therefore, we can expect some distortion in position accuracy. To account for these error sources we add noise to the simulated tracks. GPS devices typically have a minimum error which is less than 7.8 meters [2]. To model the error we add random noise defined by a two-dimensional Gaussian distribution (see Figure 5) where 65% of the measurements are between $[-2.5, 2.5]$ meters and 95% of the measurement is between $[-5.5, 5.5]$ meters around the true position.

We create each sample track with 10 way points, using the above mentioned procedure. We simulated 10 datasets with measurements every 6 seconds. After forming the track we add noise using the error model. The final dataset has 10 simulated tracks containing 82,037 data points. Here for each track we make two sets one for building the segments and the other for testing. These sets are formed by selecting the alternative points from each track data. To evaluate the accuracy we calculate the average of the root mean square error of the data points in the test set and the recovered

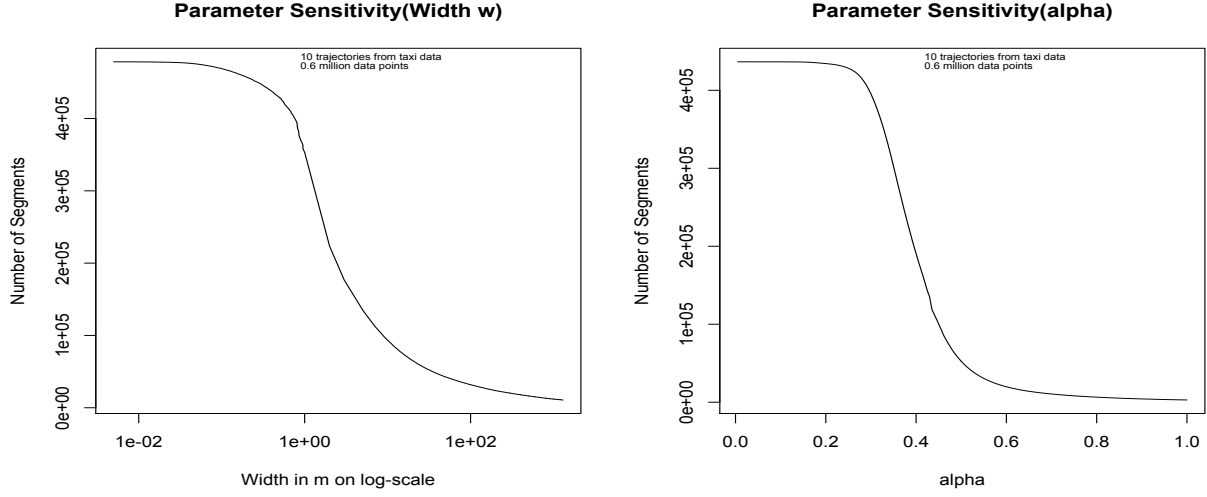


Figure 3: Parameter sensitivity for OEs $\log(w)$ and MDL (α)

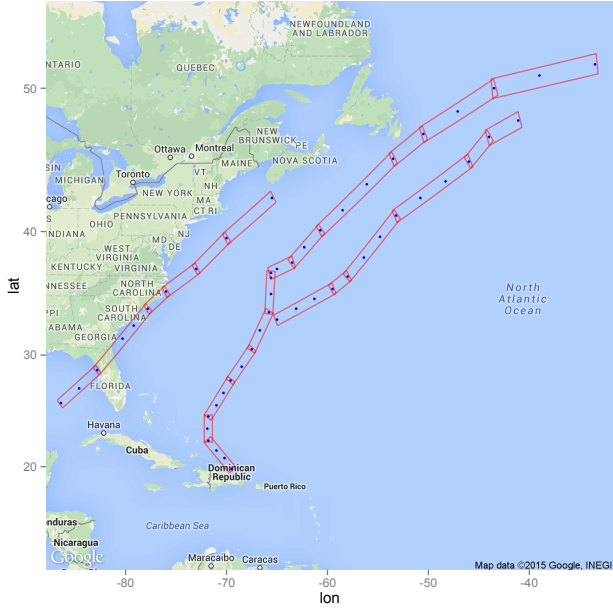


Figure 2: Three Hurricane tracks segmented with OEs (used $w = 25\text{km}$)(Note: OE's(boxes) are shown bigger for clear visualization. Some OE's width differs only due to projection.)

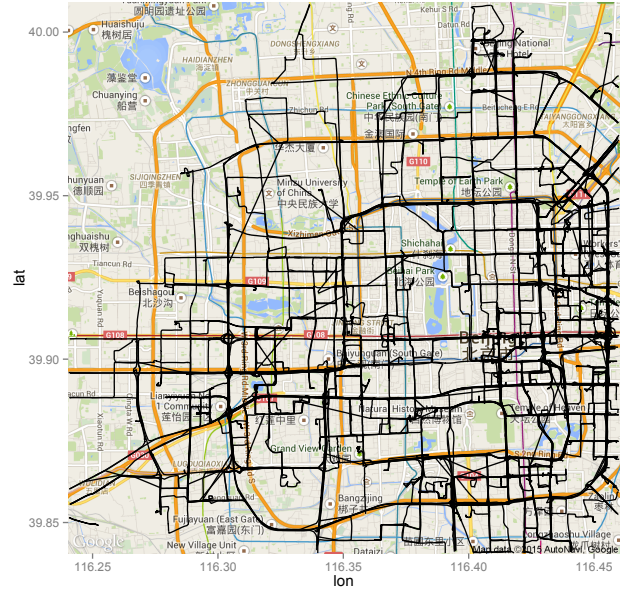


Figure 4: Taxi Dataset

track segments. Here we find the distance of the data point from the closest partitions. The average error over the 10 tracks for our proposed algorithm is shown in Figure 6 as the red line. For comparison we also show the average for the MDL-based segmentation algorithm from partition-and-detect. Figure 6 shows that the new segmentation approach outperforms the MDL-based approach for all numbers of partitions.

On the taxi data are shown in Figure 7. Figure 7 shows that the new segmentation approach also outperforms the MDL-based approach for all number of partitions for taxi data. Here the w and α are selected in a way to align the comparison of error based on the same number of segments.

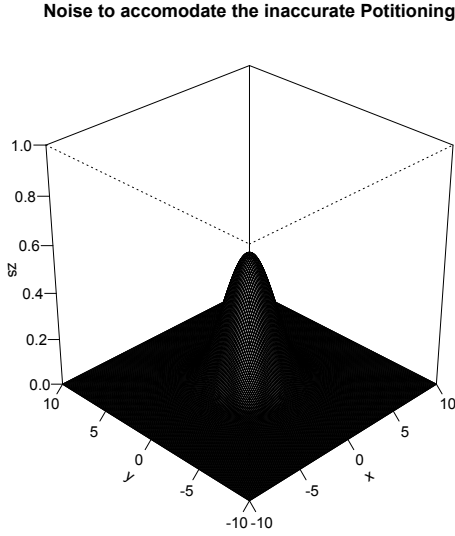


Figure 5: Error Model

In future paper we present how to tune the w without loosing the actual track. The average error for simulated tracks is more than the taxi tracks that is because of the noise added to the simulated tracks.

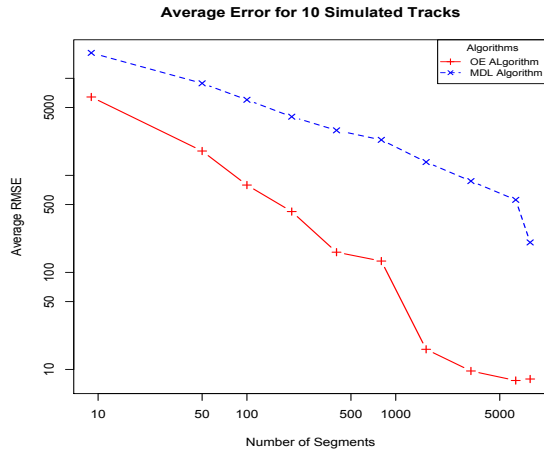


Figure 6: Error for 10 Simulated Tracks on log-log scale

5. CONCLUSIONS

In this paper we have presented a new trajectory segmentation approach based on the idea of oblique envelopes and performed experiments with hurricane data, taxi data and simulated data. We compared our algorithm with the existing partition-and-detect algorithm and showed that the new algorithm's results are much better than the existing MDL algorithm. A big advantage of the new algorithm is that its parameter w describes the maximal error for each point in

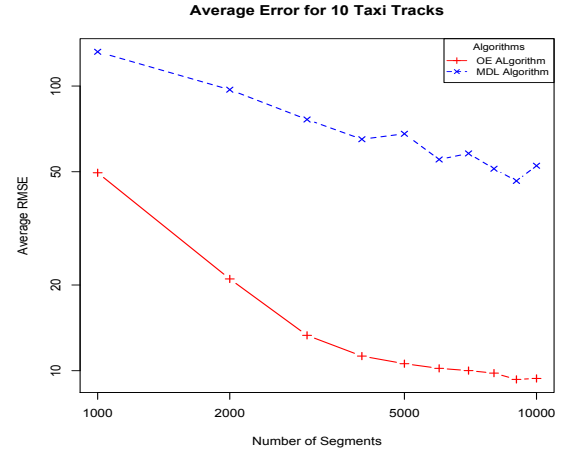


Figure 7: Error for 10 Taxi Tracks on log-log scale

a segment and is thus much easier to specify by the user.

The identified oblique envelopes can be used for many applications. In the future we will investigate their use for outlier detection, sub-trajectory clustering and destination prediction. As of now we are limited two dimensional data (Latitude and Longitude) and linear trajectories, but we extend to three dimensional (Latitude, Longitude and Altitude). We also study the curvature paths using the spline methodologies.

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